

Box Gloves – Fight Detection (How To Do)

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What is the project about and what can we achieve with this?

- The project consists in having a glove that sends information about punches (velocity / Movement) in order to help Boxing Athletes.
- With the Information that is sent will be possible to determine which type of “punch” was done.
- This is helpful for home athletes or even people that train box in their house/alone.
- This can also be used if well developed in Television showing the Force/Speed/Accuracy of the punch and be used by professionals in order to maximize their potential without wasting a lot of energy in punches.

The Workflow Process and each steps of the project should be done:

- First of all in order to “detect” what is a punch we need to create some sort of ML (machine learning) in order to know what is a punch and what type of punch was given.
- So the workflow of this process should be done like this:
 - o First step: Setup Accelerometer and Gyroscope
 - o Second step: Sent the Accelerometer and Gyroscope Information to the PC
 - o Third step: Create the initial Dataset. With This Dataset, we will be able to create a Model that will predict which type of punch was given with the Gyroscope and then we can also setup what a perfect punch is with the Speed and Movement

- Forth Step: Develop the App that shows in a 3d space the Punch in real time (or at least with low delay)
- Fifth Step : With all the data and App Working, we might be able to create a “improve punches” system App

What Electronic and which Programming Languages we will be using:

- First of all, we need a MCU. We will be using the esp32 due to the “power”/efficiency/price.
- For the Transmit/Receive Unit we will use Radio Frequency with the specific protocol of the espressif company. (esp-now)
- Battery
- Accelerometer
- Gyroscope

- C/C++
- Python (for App and 3d Real time Graphic)